

GAMIKA JAYAWARDHANA

ELECTRONICS AND COMPUTER SCIENCE UNDERGRADUATE



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Wijayasiri, Dela, Ratnapura

SUMMARY

Passionate Electronics & Computer Science undergraduate at the University of Kelaniya specializing in AI/ML Engineering and Full-Stack Development. Proven track record in architecting enterprise-grade web platforms, Agentic AI workflows, and specialized ML models alongside high-performance browser extensions. Expert in Computer Vision, Large Language Models (LLMs), and Scalable Microservices, with a unique technical foundation in Embedded Systems and UI/UX Design.

TECHNICAL SKILLS

- **AI/ML:** Machine Learning, Deep Learning, Computer Vision, PyTorch, TensorFlow, GenAI (LLMs, RAG, Prompt Engineering), Agentic AI (LangChain, LangGraph), n8n, MLOps
- **Full-Stack:** Java (Spring Boot), Python (FastAPI), JavaScript/TypeScript (Next.js, React), C# (.NET), C
- **Database & Cloud:** MySQL, MS SQL Server, MongoDB, PostgreSQL, AWS
- **Hardware/Embedded:** Arduino, ESP32-CAM, PIC, Microcontrollers, IoT, Micro C, PCB Design
- **Tools/Other:** Docker, Git, Linux, Chrome Extension Dev, UI/UX Design, Graphic Design, 3D Design

PROFESSIONAL EXPERIENCE

Founder and Freelancer, Greams

March 2023 - Present

- Leading a creative design and development service, delivering full-stack web solutions and branding for international clients.
- Architected and deployed high-performance web applications using Next.js 16 and React 19, achieving 98%+ Core Web Vital performance scores and a 40% increase in organic SEO traffic.
- Developed automated service modules and custom API integrations to streamline business operations and lead generation for global professional services.

Digital Product Promoting Assistant Trainee, Bank of Ceylon

May 2022 - Feb 2023

- Assisted in digital banking adoption and collaborated with technical teams to improve service quality and customer onboarding,

EDUCATION

Bachelor of Science (Hons) in Electronics and Computer Science

July 2023 - Present

University of Kelaniya

- Double Major in Electronics and Computer Science
- Current GPA: 3.51

Diploma in English

Jan 2021 - Dec 2022

Sabaragamuwa University of Sri Lanka

General Certificate of Education

July 2023 - Present

Sivali Central College - Ratnapura

- G.C.E. Advanced Level: Secured a District Rank of 115 with an ABC result profile.
- G.C.E. Ordinary Level: Graduated with a perfect record of 9 Distinctions (A9).

CERTIFICATIONS

- MySQL Implementation Certified Associate (Oracle Certified)
- Machine Learning I (Columbia University)
- LFS101: Introduction to Linux (The Linux Foundation)
- Introduction to FastAPI and Backend Development Fundamentals (Packt)

PROJECTS

Egocentric Assistive Navigation via Monocular Depth Estimation (On going Research Project)

PyTorch, Vision Transformers, Python, Computer Vision, ONNX

- Engineering a vision-based LiDAR alternative for assistive navigation using Monocular Depth Estimation.
- Optimising Transformer architectures for edge-device deployment to provide real-time spatial awareness.
- Curating large-scale egocentric datasets to ensure model robustness across diverse environmental transitions.

HarvestLink: Microservices-based B2B Agri Marketplace

Java (Spring Boot), React, Kafka, Docker, MySQL, Redis

- Architected distributed microservices with Kafka and Docker, ensuring scalable, event-driven communication.
- Streamlined infrastructure using an API Gateway and JWT, managing modular services for high-traffic commerce and Optimized performance for agricultural trade through reactive processing and advanced caching strategies.

AI-Powered Gmail Assistant with Browser Extension

Java, Spring Boot 4.0, React, LLMs, Gemini API

- Orchestrated a reactive backend using Spring WebFlux and Java 25, reducing API latency for LLM-generated responses by 30%.
- Built a responsive management dashboard with Vite and Tailwind CSS and Integrated the system as a Chrome Extension to provide seamless AI-assisted writing directly within the Gmail UI.

LLM-Driven WhatsApp Financial Assistant

Python (FastAPI), Groq API (Llama 3.3 70B), MongoDB, Meta WhatsApp Cloud API

- Built an AI-powered WhatsApp bot using Groq (Llama 3.3) to parse natural language into multi-action financial intents.
- Developed a resilient FastAPI/MongoDB backend featuring custom webhook deduplication and real-time dynamic balance calculations.

RideXpress – Car Rental Management

C#, .NET, MS SQL Server

- Collaborated with a University of Kelaniya team to build a desktop management system with interactive RDLC reporting.
- Designed the relational database schema and integrated secure customer/inventory management modules.

LEADERSHIP & VOLUNTEERING

○ Senior Editor Vide Club of University of Kelaniya	May 2025 – April 2026
○ Head of Designs, KelaniSTEAM Avenue Vide Club of University of Kelaniya	Dec 2024 – May 2025
○ External Relations Coordinator, Poverty and Hunger Avenue Leo Club of University of Kelaniya	Sep 2024 – May 2025